



THE UNIVERSITY OF KANSAS
Electrical Engineering & Computer Science



M.S. THESIS DEFENSE

Region-Aware Optimization of Hierarchical Bloom Filters for Privacy-Preserving Face Membership Verification

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Outline

01 Background and Related Work

02 Objective and Contributions

03 Dataset and Experimental Protocol

04 Published Baseline Analysis

05 Baseline Diagnostics and Literature

06 Landmark-Indexed Bloom Filters (DCT-LBFs and Deep Spatial LBFs)

07 Results and Conclusions

01

Background and Related Work

Face Membership Verification

TASK	QUESTION	WORK PER QUERY
Identification	Which enrolled identity matches?	1:N search + identity
Verification	Does the query match a claimed identity?	1:1 comparison
Membership verification	Is the query in the enrolled group?	1:N group test, YES / NO only

USE CASES



Renewal screening



Duplicate enrollment



Watchlist membership

237.7M U.S. licensed drivers in 2023. Large biometric galleries require fast search, compact storage and secure facial data.

Source:
FHWA

Prevailing Challenges

Accuracy

Genuine queries must survive, while strangers must not collide.

Storage

Large galleries require compact templates and efficient membership queries.

Noise Tolerance

Pose, illumination, expression and alignment change biometric codes.

Security

Stored templates should not reveal or reconstruct the original face.

Time

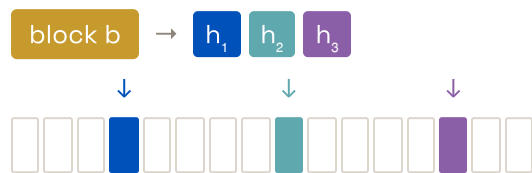
Real time processing of large volumes of data pose challenges in systems requiring immediate responses.

Face membership verification must balance accuracy, security, latency, storage and robustness.

Bloom Filters: Compact Membership Testing

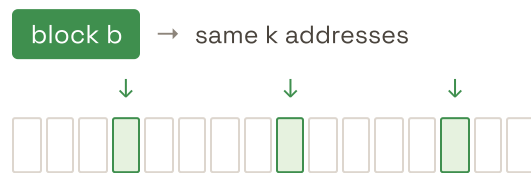
Bloom filters are compact and fast, but they provide no intrinsic tolerance to biometric variation.

(A) INSERTION



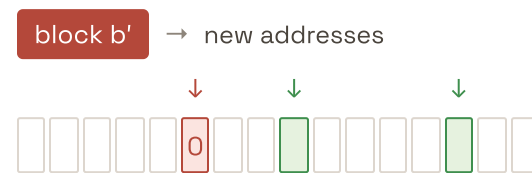
Each of the k hashes sets one bit of the m -bit array.

(B) QUERY, EXACT BLOCK



All k addressed bits are set, the block matches.

(C) QUERY, ONE BIT CHANGED



One unset bit is enough, the block is rejected.

FALSE POSITIVE

An unseen block may match by chance; f controls this probability.

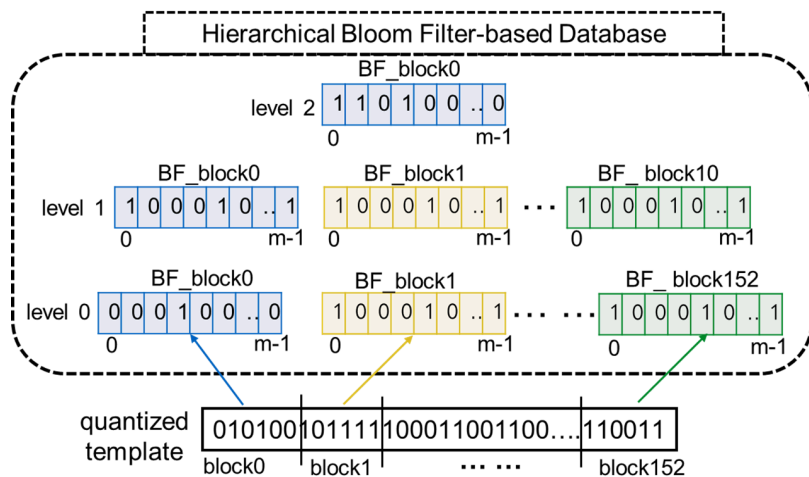
FALSE NEGATIVE

An exact inserted block is not missed under standard BF assumptions.

BIOMETRIC CAVEAT

A changed or noisy block hashes elsewhere and is rejected.

Hierarchical Bloom Filters: Enrollment and Query



(A) ENROLLMENT

Input Image

Preprocessing and Feature Extraction

Quantization

Insertion into HBF

HBF Database

Authenticated by min. x BFs?

Yes ↓
Is a Member

(B) QUERY

Query Image

Preprocessing and Feature Extraction

Quantization

Query the HBF

HBF Database

Authenticated by min. x BFs?

No ↓
Not a Member

HBF combines compact membership testing with partial agreement across facial blocks.

The Research Gap

01 **Limited evaluation under facial variation**

The published HBF shows feasibility, but was not evaluated across harder datasets

02 **Design choices were not systematically disentangled**

The effect of hierarchy, number of Bloom filters, and Bloom-filter false-positive rate was not isolated

03 **Global raw-pixel templates lack selective facial evidence**

All pixel positions are treated equally, creating long blocks that are sensitive to normal facial-image variation

04 **No controlled representation comparison**

Handcrafted features and Deep features were not compared within the same Bloom-filter membership backend.

RESEARCH QUESTION

Can Region-Aware representations improve HBF accuracy and storage efficiency?

02

Objective and Contributions

Objective

Diagnose the published HBF and develop a more accurate and compact landmark-indexed membership framework.

01

Evaluate the baseline on datasets with greater facial variation.

02

Separate hierarchy, Bloom-collision and representation limitations.

03

Identify facial regions carrying useful identity information.

04

Propose, Implement and Evaluate **Region-Aware Bloom Filters**

Thesis Contributions

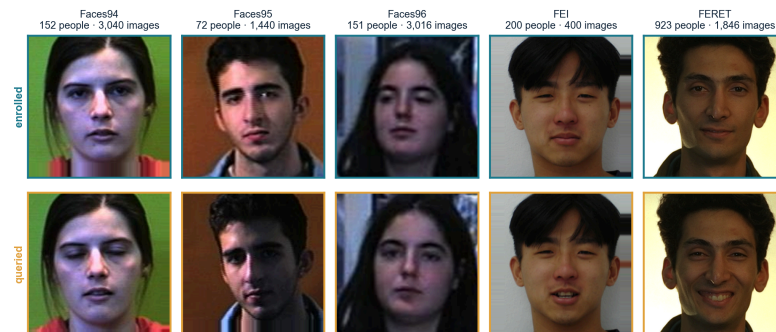
-
- 01 Reproduced and evaluated the published HBF baseline across five datasets.
 - 02 Diagnosed hierarchy, Bloom-backend, and global-representation limitations.
 - 03 Proposed LBF, a flat landmark-indexed Bloom-filter design.
 - 04 Built two representations: DCT-LBF and Deep Spatial LBF.
 - 05 Improved recognition performance while reducing template storage.
-

03

Dataset and Experimental Protocol

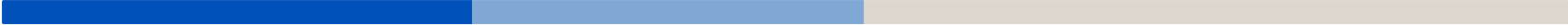
Datasets and Sources of Facial Variation

DATASET	PEOPLE	IMAGES	WHAT IT ADDS
Faces94	152	3,040	clean, consistent lighting
Faces95	72	1,440	scale, distance and lighting change
Faces96	151	3,016	head turn and cluttered background
FEI	200	400	neutral vs. smiling
FERET	923	1,846	separate capture sessions
Total	1,498	9,742	five independent sources



Identity-Disjoint Experimental Protocol

No identity crosses splits.



PARTITION	SHARE	IDENTITIES	ROLE
Fit	30%	450	Fit the mean face and representation parameters
Development	25%	375	Select landmarks, descriptors, and decision thresholds
Evaluation	45%	673	Apply frozen settings for the final untouched scores

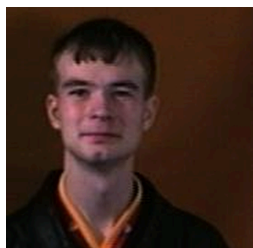
Evaluation: 503 enrolled identities · 1,625 genuine queries · 938 imposter queries

04

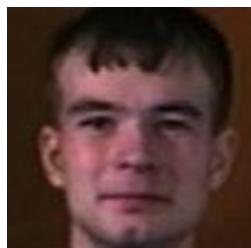
Published Baseline Analysis

Pipeline 1: Face Photo to 4,900-Bit Template

One pixel $\rightarrow$ one bit: $P = 70 \times 70 = 4,900$ bits = 613 bytes per template



01 Face photograph



02 Face detection
and eye
alignment (dlib)



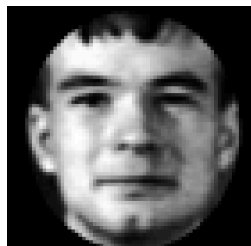
03 Resize to 160 ×
160 grayscale



04 Elliptical facial
mask



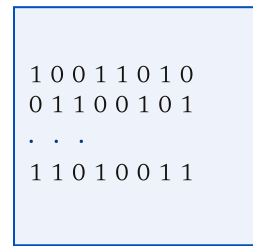
05 Histogram
equalization



06 Resize to 70 × 70



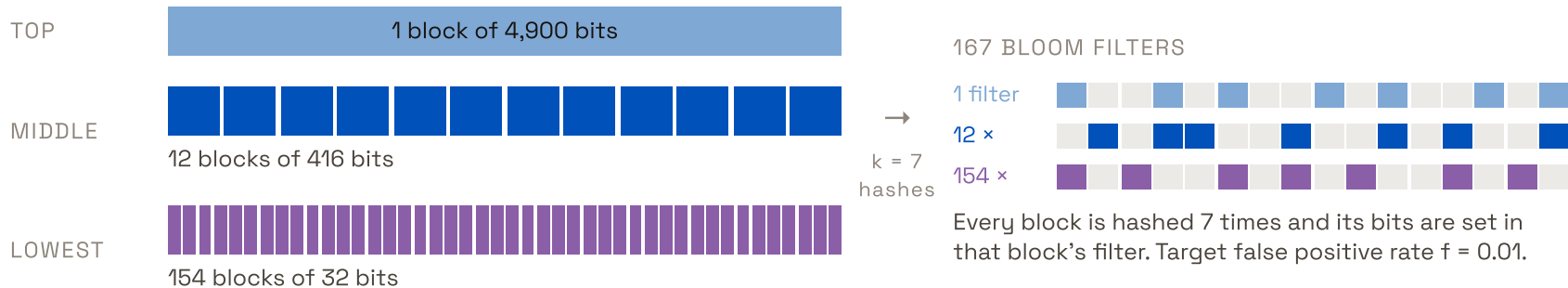
07 Compare each
pixel with the
fitted mean face



08 4,900-bit binary
template

Pipeline 1: HBF Construction

One 4,900 bit template cut into blocks, one Bloom filter per block



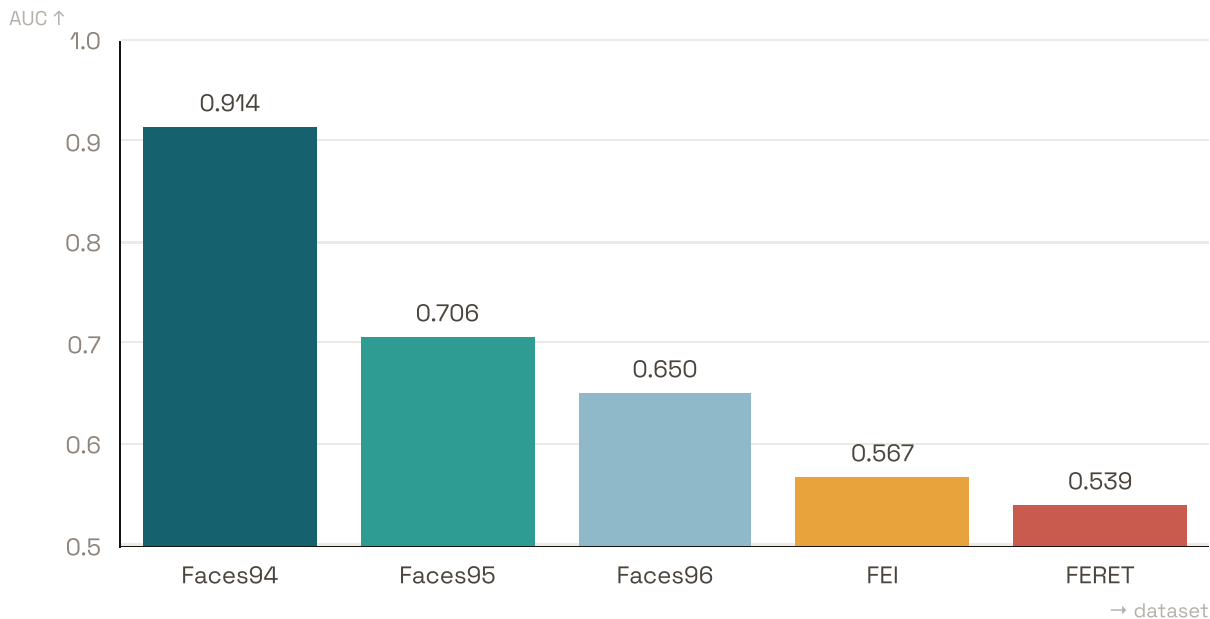
Score = how many of the 167 filters fire. A threshold turns that into member or not a member.

Pipeline 1: Recognition Performance

Strong in controlled images, weak under larger facial variation

0.6754

AUC Combined
(chance = 0.5)



Faces94 reaches 0.9140 AUC; FEI and FERET remain close to chance (0.5).

05

Baseline Diagnostics and Literature

Hierarchy-Level Contribution

BIT-FLIP RATE

$$p_t = e / P = \frac{\text{Hamming distance between two photos of the same person}}{\text{template length}}$$

PROBABILITY A BLOCK SURVIVES

$$\Pr(B_i \text{ survives}) = \left(1 - \frac{e}{P}\right)^{l_i}$$

l_i = block length in bits

LEVEL	BLOCKS	BITS/BLOCK	PREDICTED SURVIVAL	OBSERVED ROLE	AUC ALONE
Lowest	154	32	0.0190	dominant	0.639
Middle	12	416	4.19×10^{-23}	minimal	0.544
Top	1	4,900	≈ 0	none	0.499

Mean genuine bit-disagreement is $e/P = 0.1165$. The model predicts that the 416-bit and 4,900-bit levels are effectively inert. The top level is at chance

Bloom-Filter Sizing

$$m = - \frac{n \ln f}{(\ln 2)^2} \quad k = \frac{m}{n} \ln 2$$

SETTING	F	M / N	K	INTERPRETATION
Published	0.10	4.79	3	smaller filter; more collisions
This work	0.01	9.59	7	used for all three pipelines

m = Bloom-filter length in bits

n = number of elements inserted into
that Bloom filter

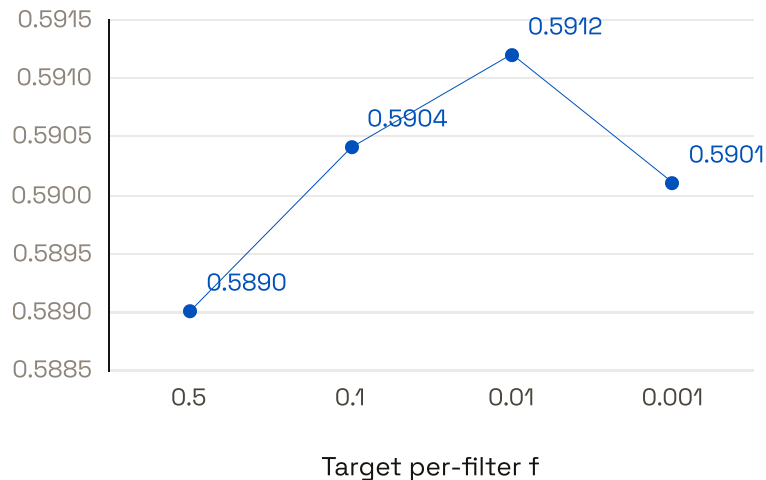
k = number of hash functions

The Backend Is Not the Bottleneck

EXPERIMENT 2A · SWEEPING F

0.0022

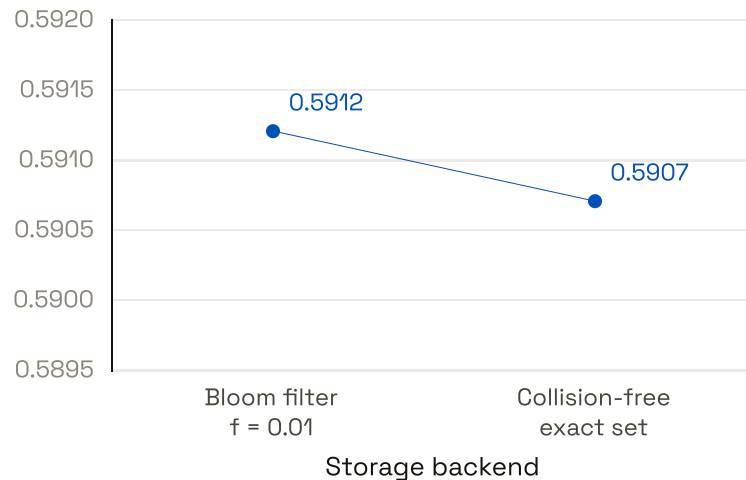
Total AUC Combined movement across the full sweep.



EXPERIMENT 2B · EXACT SET

0.0005

Removing Bloom collisions entirely moves AUC 0.5912 → 0.5907.



Both controls reach the same ceiling: the representation, not the filter, limits recognition.

Binary-Template Information

Representative aligned Faces94 image
within the baseline mask



1,635 of 4,900 positions (shaded)
are constant across the gallery



33.4%

of positions, 1,635 of 4,900, are constant across the gallery.

12

complete 32-bit blocks are constant, so they fire for nearly everyone.

The raw global template spends storage on common face structure rather than identity variation.

Do Some Facial Regions Carry More Identity Information?

Sadr et al. (2003) [7]

Removing eyebrows impaired recognition more than removing the eyes.

Abudarham & Yovel (2016) [9]

Identity information is unequal across features, and the critical ones stay stable across views.

Karczmarek et al. (2014) [8]

The eye–eyebrow area was the most descriptive region; eyebrows alone exceeded 81% accuracy.

Lestriandoko et al. (2022) [10] *

For ArcFace, eyebrows ranked most informative, then eyes, nose, mouth.

Identity evidence is uneven, and the periocular region carries the most of it.

* Measured on a system trained with the ArcFace loss, the closest published setting to Pipeline 3.

06

Landmark-Indexed Bloom Filters (LBFs)

LBFs : Design Requirements

Remove the hierarchy

Keep one flat level; upper HBF levels add little recognition value.

Replace wasted template space

Reclaim constant bits and uninformative blocks for useful local evidence.

Index filters by facial location

One filter represents one anatomical region, not an arbitrary global bit block.

Keep the Bloom backend

The filter sizing and hashing are sound; the main gain is in the representation.

Prioritize informative face regions

Identity evidence is uneven; brows and eyes contribute more than many other regions.

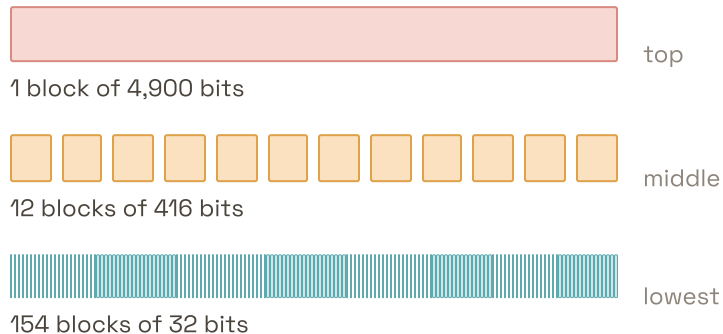
Test two local representations

Use the same landmark-indexed Bloom backend with DCT-LBF and Deep Spatial LBF.

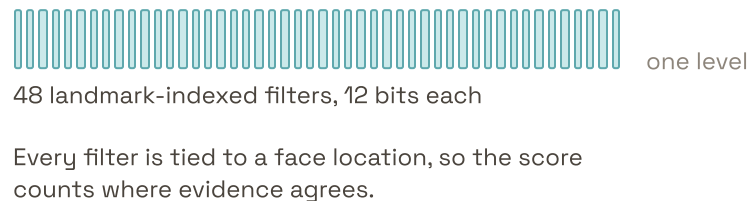
Backend: HBF vs. LBF

Same Bloom backend, hashing algorithms and decision rule; what changes is how the bits are partitioned

HBF · THREE NESTED LEVELS



LBF · ONE FLAT, LANDMARK-INDEXED LEVEL



167 position-blind filters over 4,900 raw bits become 48 filters, each anchored to a facial landmark.

06 · LBF

DCT-LBF

The DCT of One Patch

Introduced by Ahmed, Natarajan and Rao in 1974 [4]. Became core part of JPEG compression.

The DCT changes an patch from pixel values into coarse-to-fine cosine pattern measurements.

$$C_r(u, v) = \alpha(u)\alpha(v) \sum_{a=0}^7 \sum_{b=0}^7 \tilde{I}_r(a, b) \cos\left(\frac{\pi(2a+1)u}{16}\right) \cos\left(\frac{\pi(2b+1)v}{16}\right)$$

a, b

Patch positions, each from 0 to 7

v

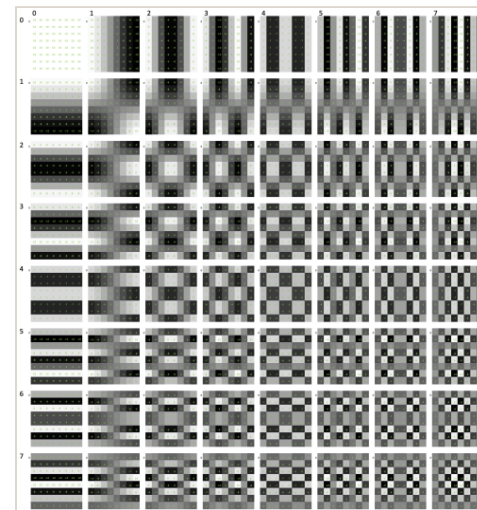
Vertical pattern frequency

u

Horizontal pattern frequency

$C_r(u, v)$

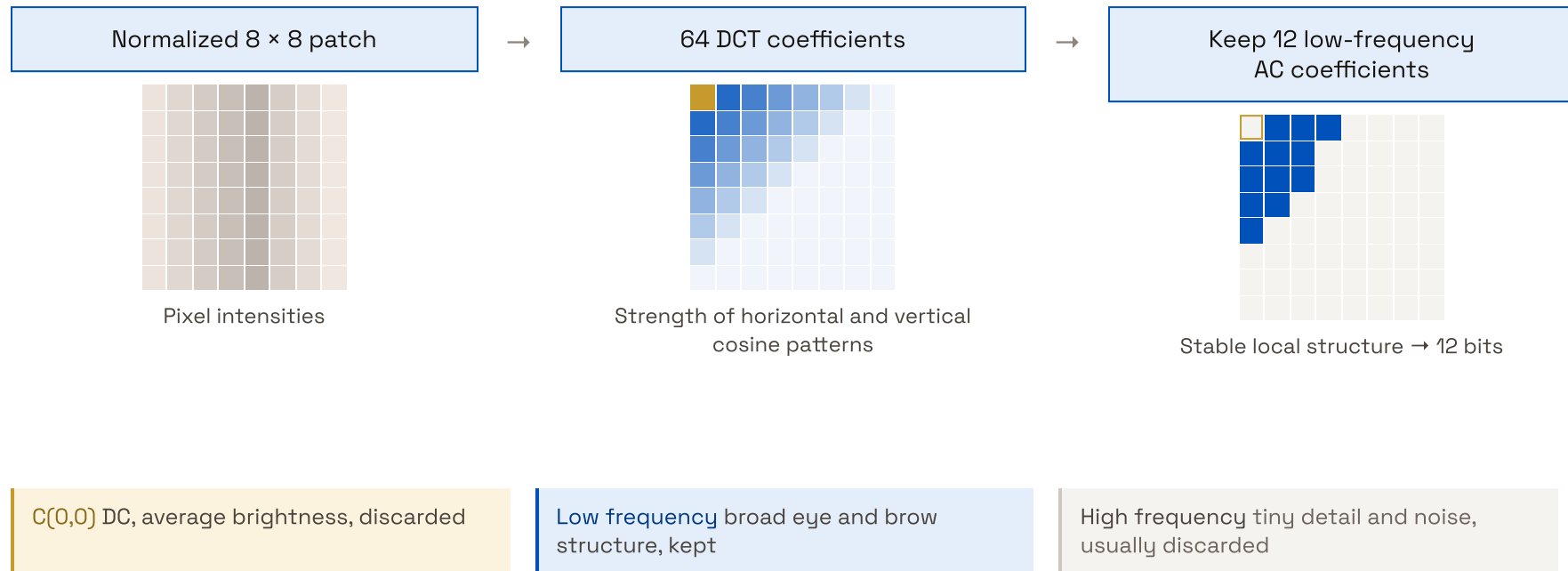
Strength of fixed pattern (u, v) in that patch



The 64 fixed cosine patterns, u across and v down [11]

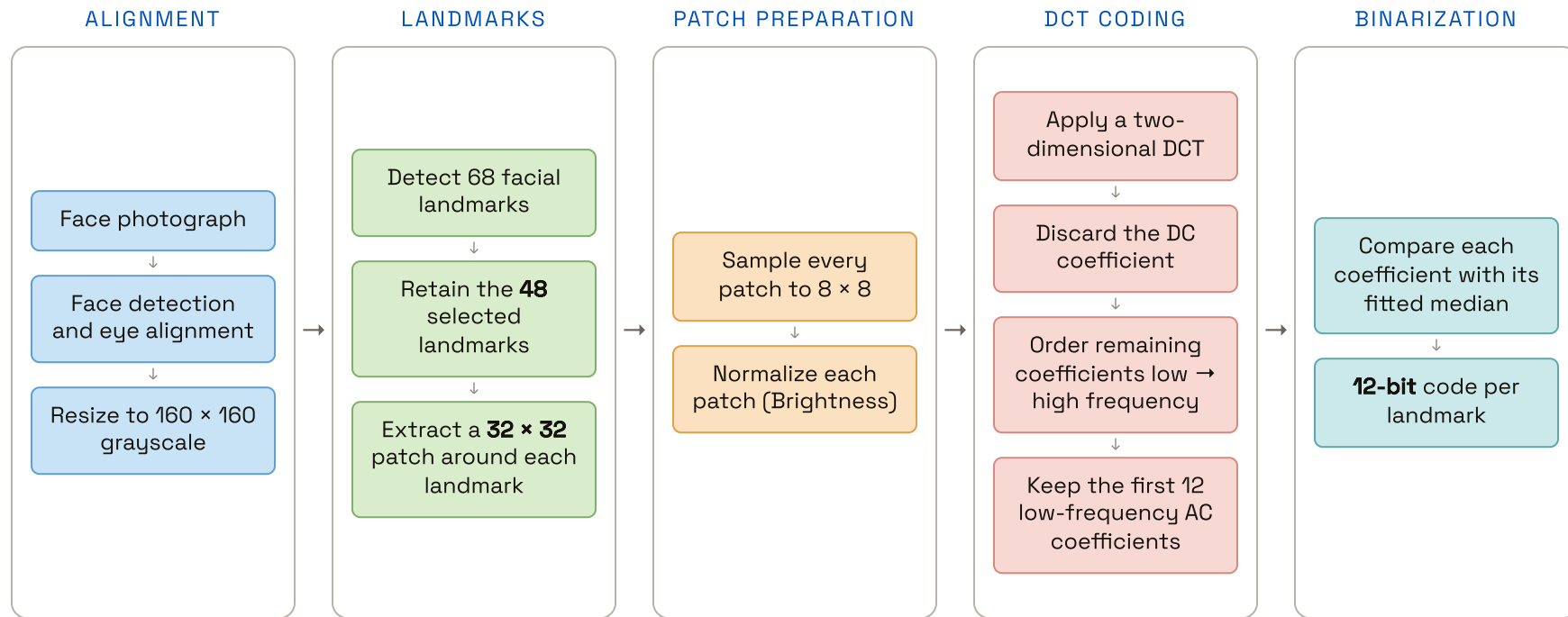
DCT: Discrete Cosine Transform

There are 64 standard frequency patterns in 2D DCT. Every coefficient is a weighted sum over all 64 pixels, weighted by one fixed cosine pattern.



Pipeline 2: DCT-LBF Architecture

Local DCT codes replace global raw pixels; the Bloom backend is unchanged



Fit: sets medians for the first 12 low-frequency AC positions

Development: selects the 48 landmark locations, bit length, and patch size

Evaluation: stays untouched

Why DCT for Local Facial Regions?

Same faces, landmarks, filters, and 12-bit budget, only the descriptor changes

Compact local structure

It summarizes a normalized 8×8 region instead of preserving raw pixels.

Coarse before fine

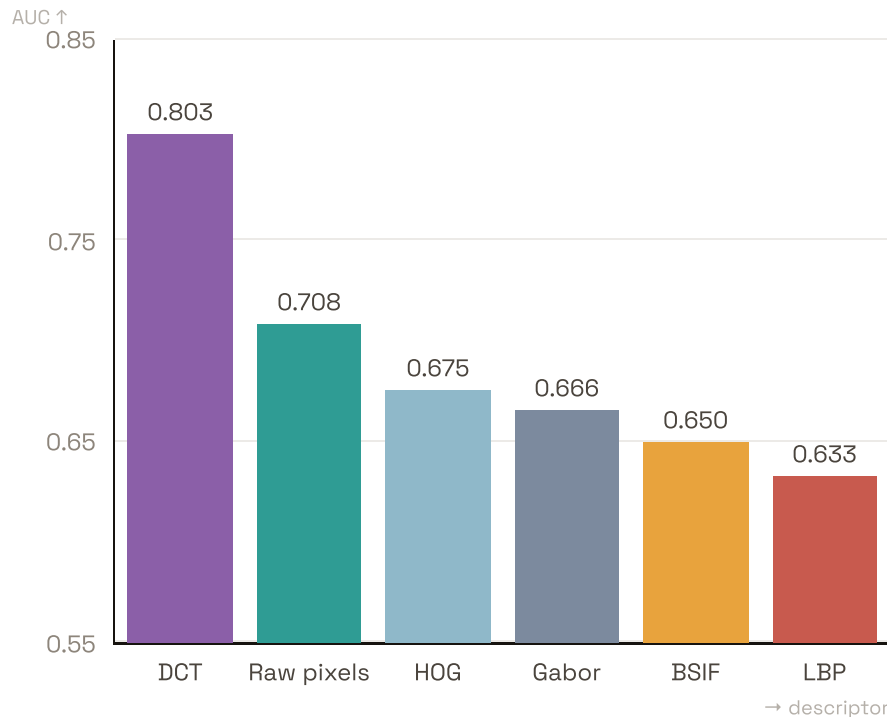
Coefficients arrive in frequency order, so a fixed prefix keeps the broad structure of the patch.

Stable and discriminative

Low-frequency structure is more repeatable across photos than fine detail.

First under a controlled swap

DCT leads every alternative descriptor on all five datasets and on the pooled bank.



Pooled AUC Combined by descriptor, 12-bit codes

Rationale for the DCT Frontend

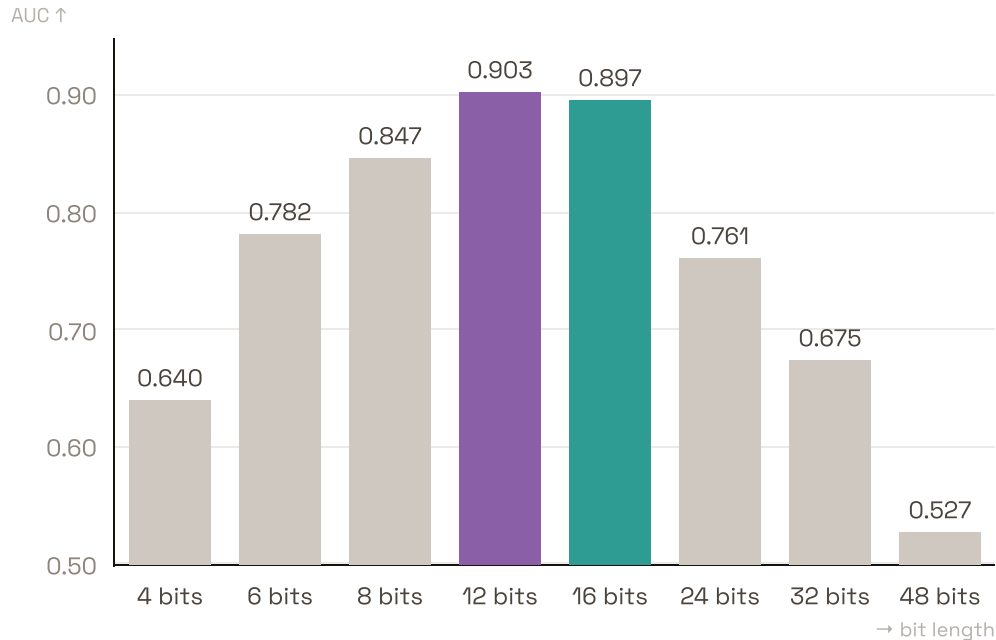
A useful Bloom code must be stable for genuine queries and rare for imposters

DESCRIPTOR	BIT-FLIP RATE	IMPOSTER FIRING	CODE VARIETY	BIT CORRELATION	VERDICT
DCT	0.251	0.105	0.978	0.171	good at both
Raw pixels	0.222	0.328	0.879	0.310	stable, not rare
Gabor	0.247	0.344	0.888	0.327	stable, not rare
BSIF	0.309	0.071	0.988	0.144	rare, not stable
LBP	0.292	0.128	0.973	0.181	rare, not stable

DCT decorrelates the local patch, reducing redundant bits and imposter matches.

Sweeps for Local Geometry and Bit Length

Sampling dimensions and code length were chosen in development, then frozen



A local 32×32 window, standardized and sampled to 8×8 : enough context for structure, not too global.

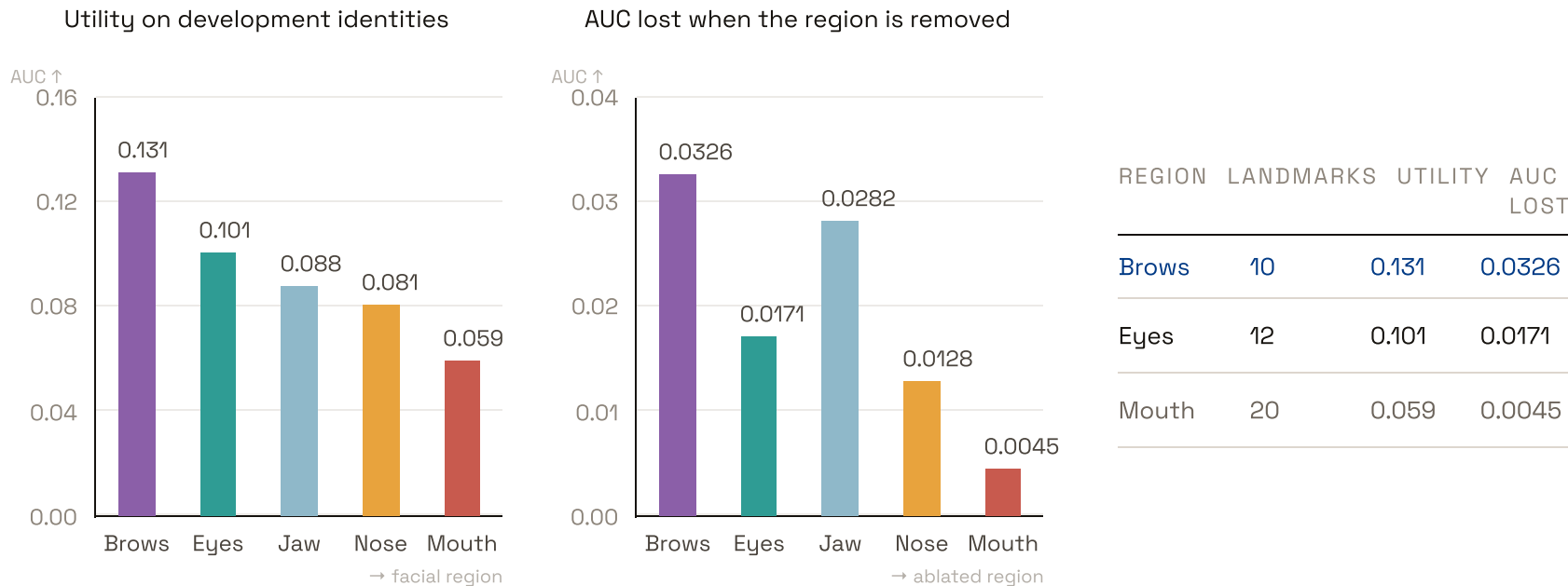
Short blocks (4–8 bits) increase content collisions; long blocks (24–48) are less repeatable across photos.

12 bits is the strict development AUC Combined winner: 0.903 against 0.897 at 16 bits.

Decision: $32 \times 32 \rightarrow 8 \times 8$, 12 bits per retained landmark. The sweep selects code length, not a per-landmark coefficient window.

Pipeline 2: Facial-Region Utility and Ablation

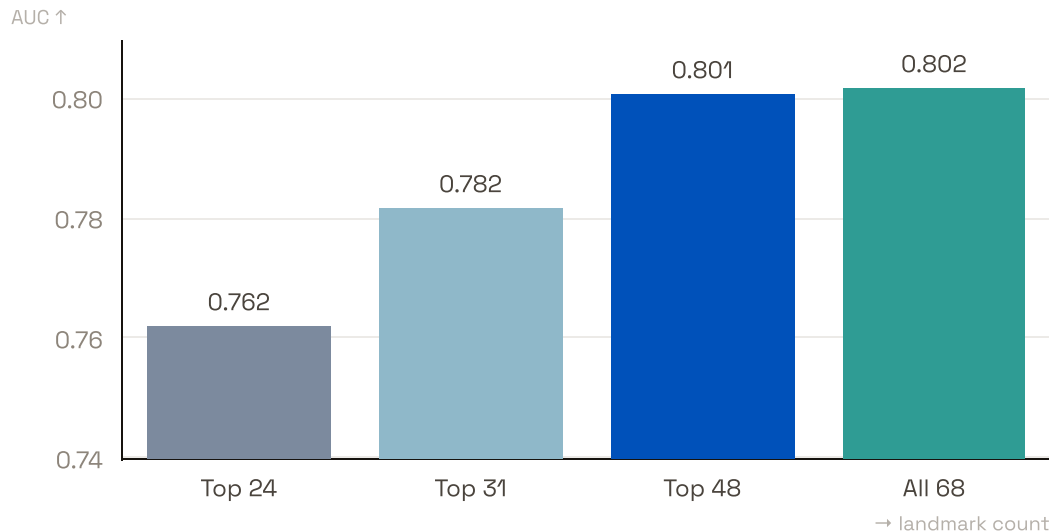
Two complementary development checks, does a region discriminate alone, and what is lost without it



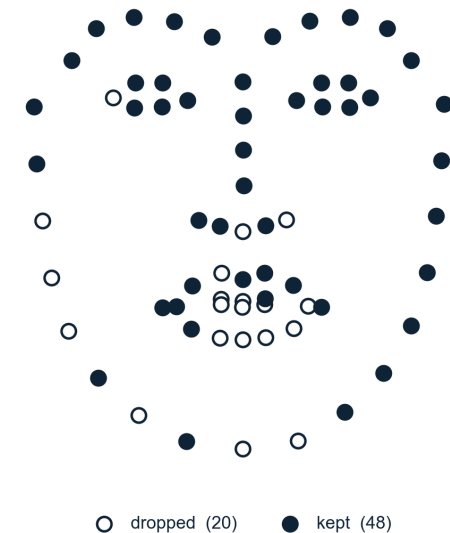
Both checks agree: brows and eyes deserve priority. This selects landmark locations only; the 12-bit DCT prefix is fixed.

Why 48 Landmarks?

Rank all 68 candidates by **development-only utility**, then choose the compact knee



Development AUC Combined by number of landmarks kept



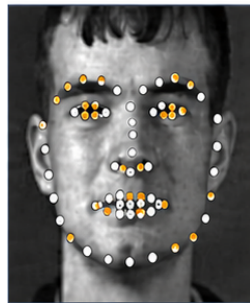
Top 48 reaches 0.801 against the 0.802 all-68 reference, 20 fewer filters for a 0.001 trade-off. Below about 30 landmarks, performance falls quickly.

Pipeline 2: One-Region Walkthrough

$32 \times 32 \rightarrow 8 \times 8 \cdot 12 \text{ bits} \cdot 48 \text{ landmark locations} \cdot \text{development-selected threshold}$



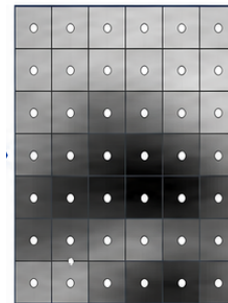
01 Preprocessing



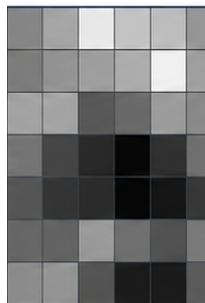
02 Localization



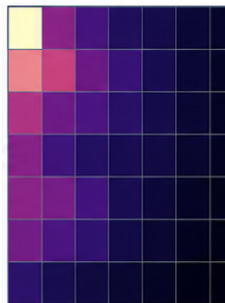
03 Sampling



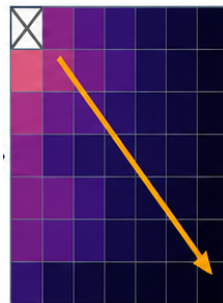
04 Resampling



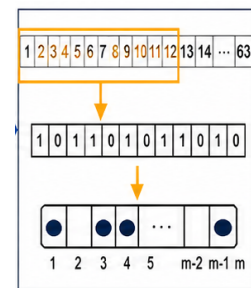
05 Normalization



06 Transformation



07 Frequency ordering

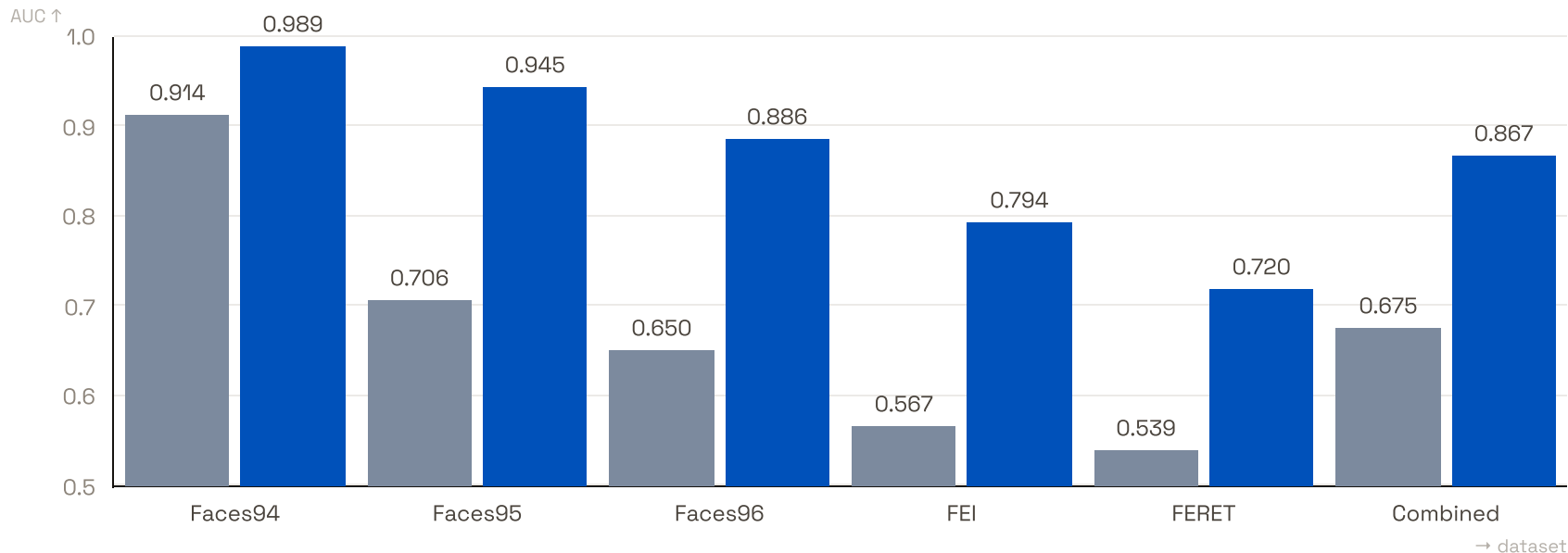


08 Fixed encoding 12 bits

48 filters, 12 bit BFs, a 72-byte template.

Pipeline 2: Recognition Performance

AUC Combined 0.6754 → 0.8668 with a 72-byte template and 48 filters



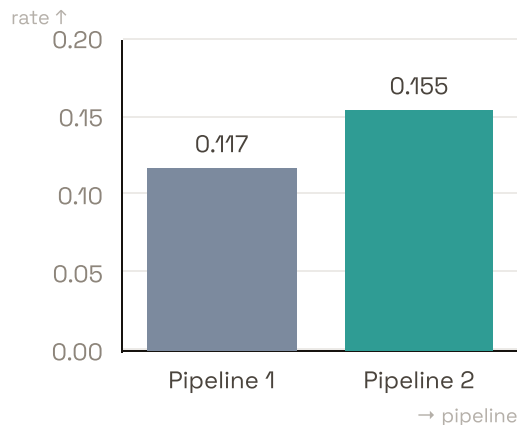
■ Pipeline 1 ■ Pipeline 2

Gains of +0.075 to +0.239 per dataset; +0.1915 combined.

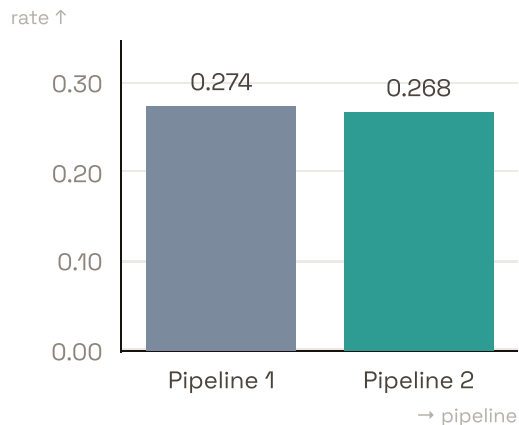
Pipeline 2: Sensitivity and Selectivity

Selectivity improves sharply; exact genuine-code survival stays near Pipeline 1

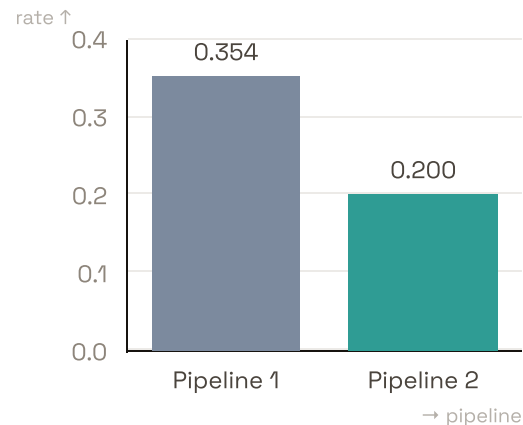
Same-person bit-change sensitivity
(lower is better)



Exact genuine-code survival
(higher is better)



Impostor collision
(lower is better)



Same-person sensitivity e/P 0.1165 → 0.1551

Genuine-code survival 0.2736 → 0.2677

Imposter collision 0.3545 → 0.1997

The remaining challenge is photo sensitivity inside the local code. This motivates the learned Pipeline 3 frontend.

06 · LBF

Deep Spatial-LBF

Pipeline 3: Motivation

Identity-trained local features target the remaining Pipeline 2 recognition gap

01 Address the remaining failures

Pipeline 2 still struggles with difficult lighting, scale, pose, and expression changes.

02 Improve genuine-code stability

Preserve Pipeline 2's impostor selectivity while making same-identity codes more consistent

03 Use identity-trained features

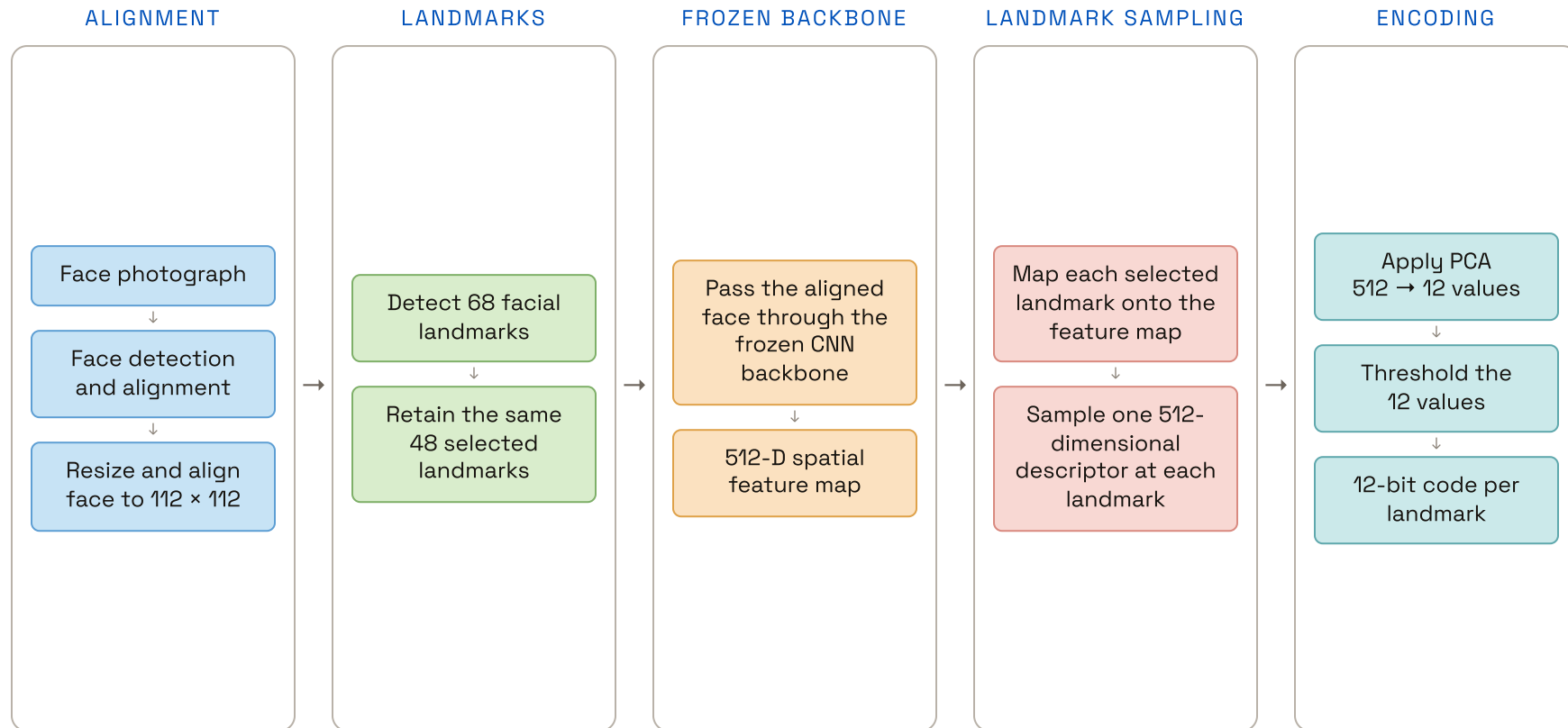
Replace handcrafted DCT appearance with frozen CNN features trained to suppress noise variation

04 Keep the comparison controlled

Change only the descriptor frontend; retain the same 48 landmarks, 12 bits per landmark, 72-byte template, HBF backend, and evaluation protocol.

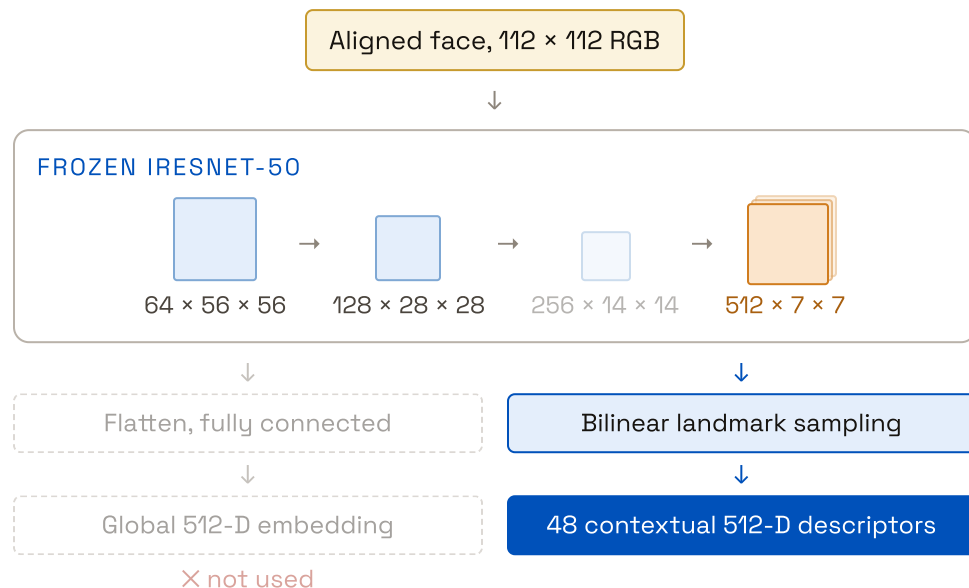
Pipeline 3: Deep Spatial LBF Architecture

Landmark-indexed Bloom filters over CNN spatial features

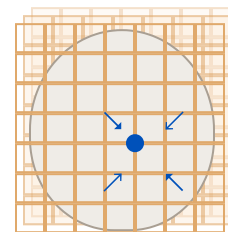


The CNN Spatial Feature Map

What the network holds at each face location, before it is pooled into one embedding.
The descriptor is local in where it is sampled, but contextual in what it knows.



SAMPLING ONE LANDMARK



A landmark rarely lands on a cell centre, so its descriptor is interpolated from the four neighbouring cells, across all 512 channels.

Why PCA (Principal Component Analysis)?

01 Remove redundancy

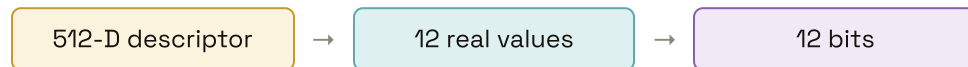
The CNN gives a 512-dimensional descriptor, but many dimensions are correlated. PCA concentrates its dominant variation into 12 directions.

02 Produce balanced binary codes

PCA centres and rescales the components. Thresholding at zero then produces balanced sign bits.

03 Match Pipeline 2 fairly

We retain 12 bits per landmark, so both pipelines store $48 \times 12 = 576$ bits = 72 bytes.



FRONTEND CONFIGURATION

BACKBONE

Frozen IResNet-50, no fine-tuning

TRAINING DATASET

WebFace600K

TRAINING OBJECTIVE

ArcFace loss

INPUT

112 × 112 RGB face

OUR EXTRACTION POINT

Final stage, $512 \times 7 \times 7$

OUR MODIFICATION

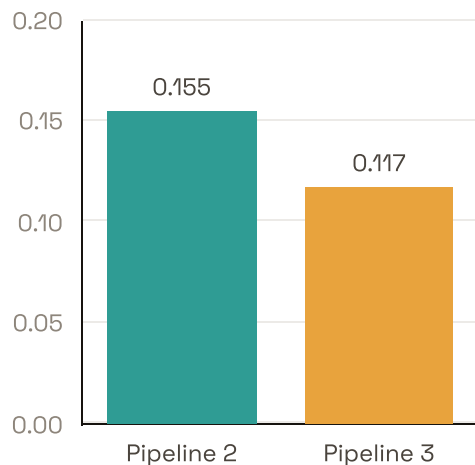
Bilinearly sample the spatial map at 48 selected landmarks

PCA converts a large, redundant CNN descriptor into a compact and balanced representation while preserving the same 72-byte template budget as Pipeline 2.

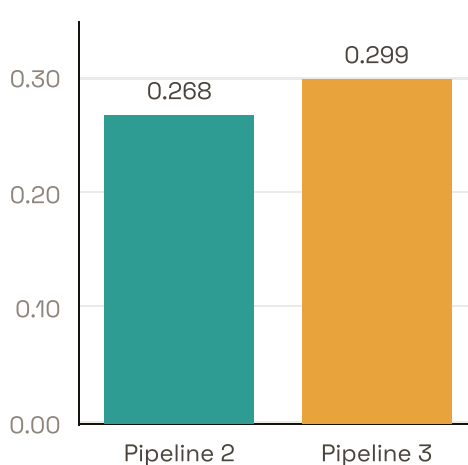
Pipeline 2 vs Pipeline 3: Sensitivity and Selectivity

Same backend and 72-byte template; only the local representation changes

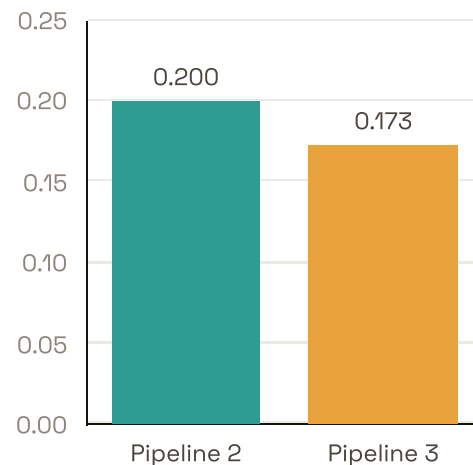
Same-person bit-change sensitivity
(lower is better)



Exact genuine-code survival
(higher is better)



Impostor collision
(lower is better)



Same-person sensitivity e/P 0.1551 → 0.1175

Genuine-code survival 0.2677 → 0.2992

Imposter collision 0.1997 → 0.1727

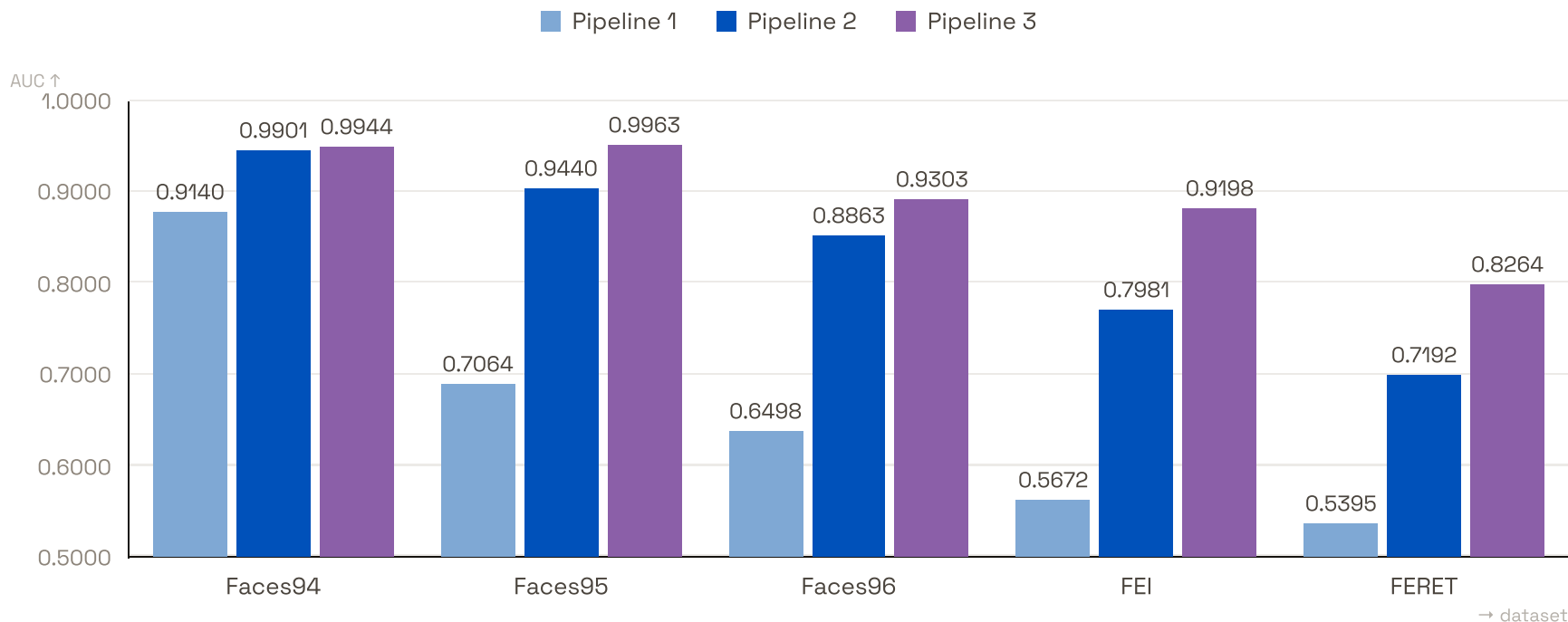
Pipeline 3 is more stable for genuine queries and more selective against strangers.

07

Results and Conclusions

All Pipelines: Per-Dataset AUC

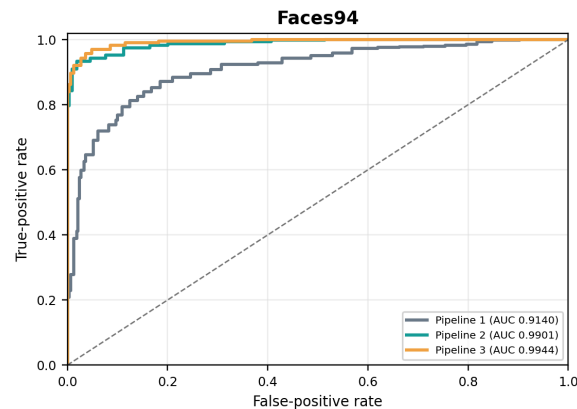
Pipeline 3 leads on every dataset



AUC Combined: 0.6754, 0.8675, 0.9334. Combined accuracy rises from 0.6751 to 0.7957 to 0.8631.

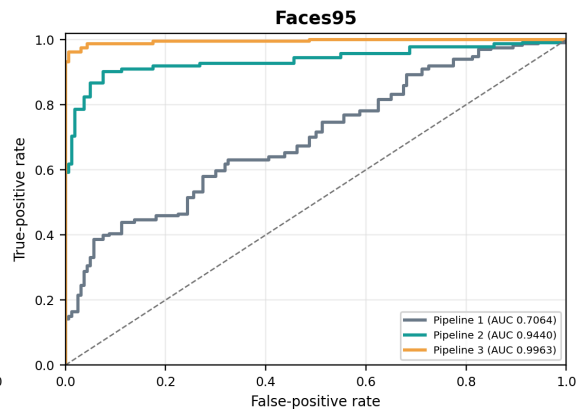
ROC Curves: Faces94, Faces95, Faces96

True-positive rate against false-positive rate



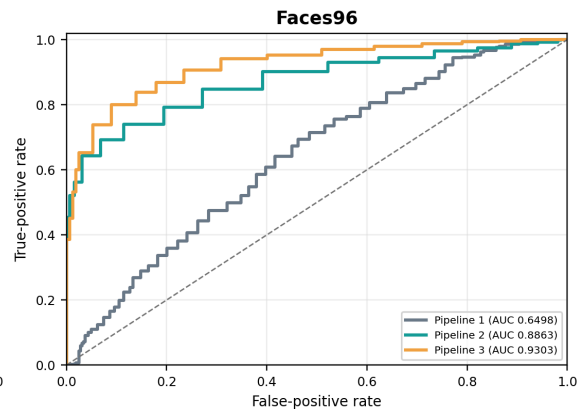
Faces94

P1 0.9140 P2 0.9901 P3 0.9944



Faces95

P1 0.7064 P2 0.9440 P3 0.9963

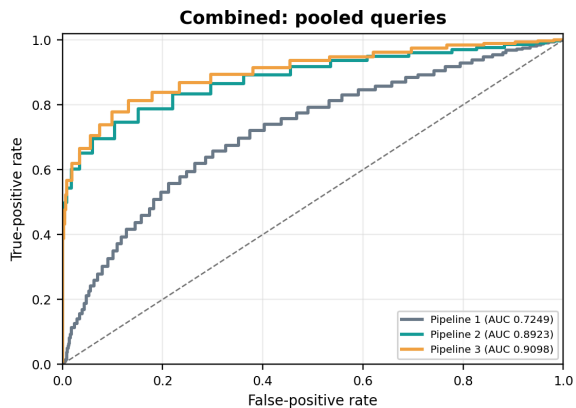
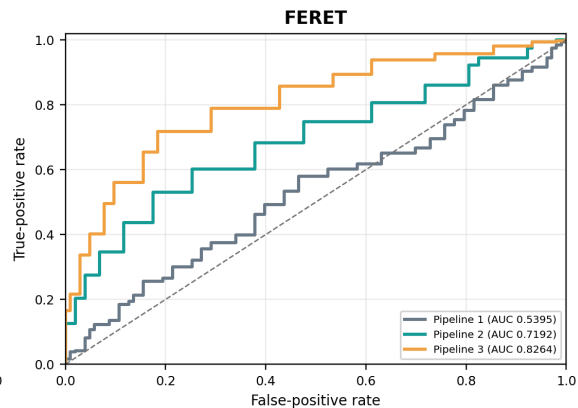
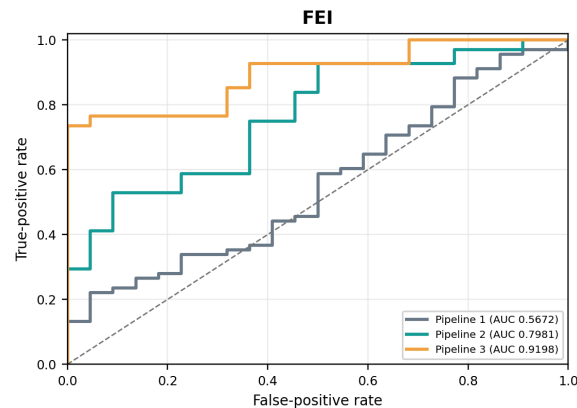


Faces96

P1 0.6498 P2 0.8863 P3 0.9303

ROC Curves: FEI, FERET, Combined

True-positive rate against false-positive rate



FEI

P1 0.5672 P2 0.7981 P3 0.9198

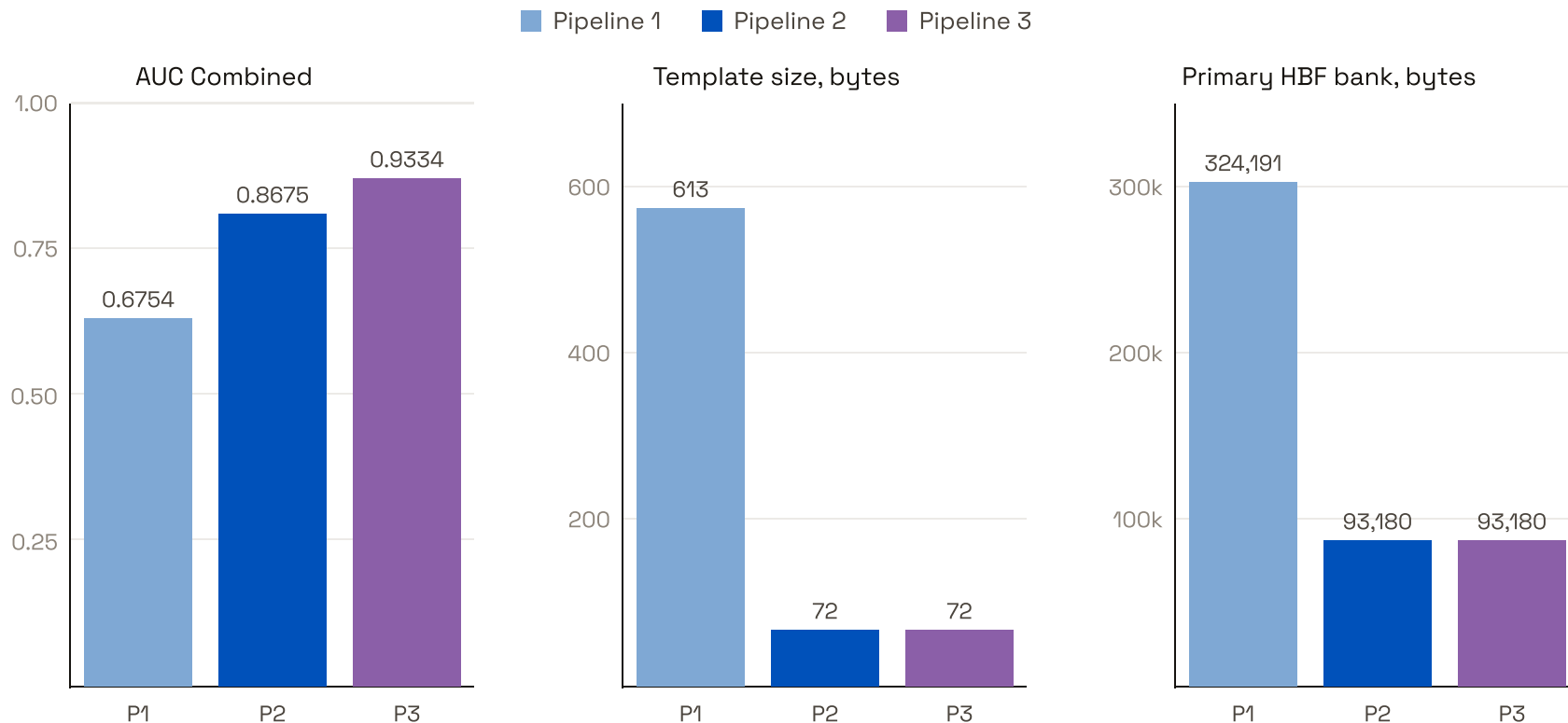
FERET

P1 0.5395 P2 0.7192 P3 0.8264

Combined

P1 0.6754 P2 0.8675 P3 0.9334

Overall Pipeline Comparison



AUC Combined rises from 0.6754 to 0.8675 to 0.9334. P2 and P3 reduce template size by 88.3% and HBF bank size by 71.3%.

Research Contributions

01 Analysis on harder datasets

Reproduced the published HBF baseline across five datasets with identity-disjoint evaluation.

02 Diagnosis of Published Baseline

The global raw-pixel representation, not Bloom-filter sizing limits the baseline.

03 Landmark-indexed Bloom filters

Replace three weak hierarchy levels with one flat level tied to facial locations.

04 DCT-LBF

Local DCT codes raise AUC from 0.6754 to 0.8675 at 72 bytes per template.

05 Deep Spatial LBF

ArcFace spatial codes reach 0.9334 AUC with the same 72-byte template.

06 Main result

Recognition improves while template storage drops 88.3% and bank storage drops 71.3%.

Future Works

01 Deployment and large-scale evaluation

02 Occluded Facial Membership

03 End-to-end storage and latency

04 Improved regional patches & Landmark selection

05 Region-specific deep encoders

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Questions and Discussion